

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of Cyber Security

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CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

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PROJECT TITLE

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OUTLINE

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- Objective
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ABSTRACT

CyberSage AI is a rule-based Expert System developed to assist users and security analysts in identifying potential cyber threats through logical reasoning. Unlike traditional antivirus software that relies on signature databases or Machine Learning techniques, CyberSage AI uses Classical Artificial Intelligence concepts such as Knowledge Representation, Forward Chaining, Backward Chaining, and Propositional Logic to analyze security events and provide intelligent recommendations. The system evaluates suspicious activities such as repeated failed login attempts, unauthorized USB connections, firewall status changes, and unknown application execution. Based on predefined logical rules stored in its knowledge base, the AI determines the most probable cyber threat, explains the reasoning behind its decision, and recommends appropriate response actions. This transparent decision-making process helps users understand why a particular conclusion was reached while improving cybersecurity awareness. The proposed system demonstrates the practical application of Classical AI techniques taught in the Foundations of Artificial Intelligence course and provides an easy-to-use, educational, and reliable decision support tool for cybersecurity.

OBJECTIVE

To develop a rule-based Expert System for identifying potential cybersecurity threats.

To implement Forward Chaining and Backward Chaining algorithms for logical decision-making.

To represent cybersecurity knowledge using a Knowledge Base with IF–THEN rules.

To analyze security events and identify possible cyber threats based on predefined logical rules.

To provide clear explanations for every decision made by the AI, ensuring transparency and user understanding.

To recommend appropriate response and preventive measures for the identified cyber threats.

To demonstrate the practical application of Classical Artificial Intelligence concepts in cybersecurity without using Machine Learning or Deep Learning.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
Rule-Based Expert Systems: The MYCIN Experiments of the Stanford Heuristic Programming Project	1984	B. G. Buchanan & E. H. Shortliffe	Rule-based knowledge representation; inference rules; explanation
Towards a Taxonomy of Intrusion-Detection Systems	1999	H. Debar, M. Dacier & A. Wespi	Intrusion detection taxonomy; misuse/anomaly detection
Designing a Fuzzy Rule Based Expert System for Cyber Security	2012	Kerim Goztepe	Fuzzy IF–THEN rules; rule repository; expert reasoning
Expert System for Intrusion Detection and Its Applications: A Brief Overview	2021	E. Kesavulu Reddy	Rule-based expert systems; misuse detection; AI-based IDS
A Rule-Based Expert System for Cybersecurity Threat Detection: Evolution, Applications, and the Hybrid AI Paradigm	2025	Abdallah Quffa & Samy S. Abu-Naser	Rule-based threat detection; explainability; hybrid AI comparison

EXISTING SYSTEM

Most existing cybersecurity solutions rely on **signature-based detection**, which can only identify known threats.

Many security tools use **Machine Learning or Deep Learning**, making them complex and outside the scope of classical AI.

Existing systems often provide **limited explanation** about how a threat was identified, making their decisions difficult to understand.

Enterprise cybersecurity platforms are **expensive** and require skilled professionals to configure and operate.

Traditional security tools mainly **detect threats** but provide limited guidance on the appropriate response or preventive actions.

Existing systems require **frequent updates** to their signature databases to detect newly discovered threats.

Most tools are designed for professionals and are **not beginner-friendly** for students or users learning cybersecurity concepts.

PROPOSED SYSTEM

• Develop a Rule-Based Expert System to identify potential cybersecurity threats using Classical AI techniques.

• Utilize Forward Chaining and Backward Chaining algorithms to perform logical reasoning based on predefined IF–THEN rules.

• Store cybersecurity knowledge in a structured Knowledge Base to analyze suspicious security events effectively.

• Provide explainable AI decisions by showing the logical reasoning behind every identified cyber threat.

• Recommend appropriate incident response and preventive measures based on the detected threat.

• Offer a user-friendly interface that allows users to enter security events and receive clear cybersecurity guidance.

• Improve cybersecurity awareness by helping users understand different attack scenarios through logical explanations.

• Demonstrate the practical implementation of Knowledge Representation, Propositional Logic, and Expert Systems from the Foundations of Artificial Intelligence syllabus

Novelty in Proposed System

• Uses Classical Artificial Intelligence techniques instead of Machine Learning or Deep Learning, making the system simple, transparent, and syllabus-oriented.

• Implements an Explainable Expert System that clearly shows the logical reasoning behind every cybersecurity decision.

• Combines Knowledge Representation, Forward Chaining, and Backward Chaining to analyze cyber threats effectively.

• Recommends immediate response actions along with threat identification, helping users take appropriate security measures.

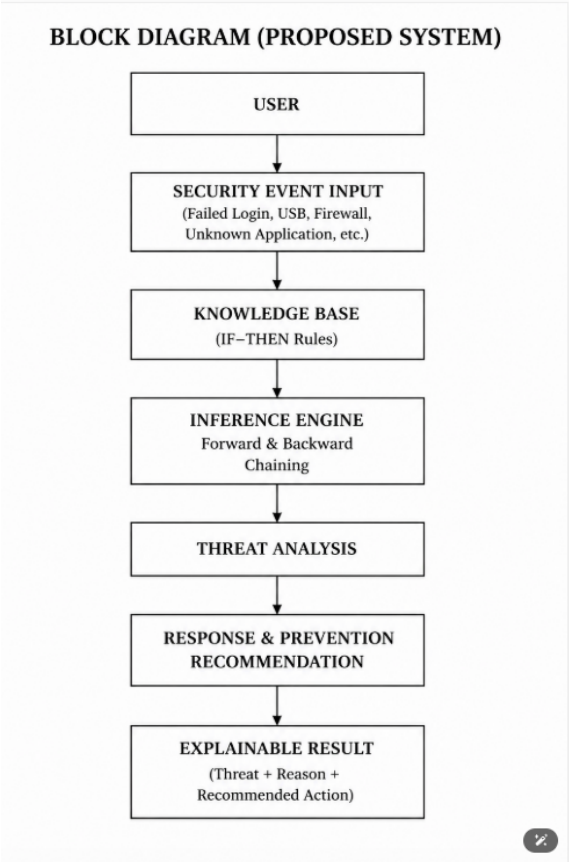
• Designed as an educational cybersecurity assistant, enabling students and beginners to understand cyber threats through logical explanations.

• Provides a user-friendly interface that simplifies cybersecurity analysis without requiring advanced technical knowledge.

• Can be easily expanded by adding new cybersecurity rules to the knowledge base without changing the inference engine.

• Demonstrates the practical application of Classical AI concepts from the Foundations of Artificial Intelligence syllabus in solving real-world cybersecurity problems.

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirement:

- Processor : Intel Core i3 / AMD Ryzen 3 or above
- RAM : Minimum 4 GB (Recommended 8 GB)
- Storage : 500 MB free disk space
- Display : 1366 × 768 resolution or higher
- Key & Mouse

Software Requirement:

s • Operating System : Windows 10/11 or Linux

- Programming Language : Python 3.10+
- IDE : Visual Studio Code
- Web Framework : Flask
- Frontend : HTML, CSS, JavaScript
- Database : SQLite (or JSON-based Knowledge Base)
- Web Browser : Google Chrome / Microsoft Edge

References

[1] B. G. Buchanan and E. H. Shortliffe, *Rule-Based Expert Systems: The MYCIN Experiments of the Stanford Heuristic Programming Project*, Addison-Wesley, 1984.

[2] H. Debar, M. Dacier and A. Wespi, “Towards a Taxonomy of Intrusion-Detection Systems,” *Computer Networks*, vol. 31, no. 8, pp. 805–822, 1999.

[3] K. Goztepe, “Designing a Fuzzy Rule Based Expert System for Cyber Security,” *International Journal of Information Security Science*, vol. 1, no. 1, pp. 13–19, 2012.

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[5] A. Quffa and S. S. Abu-Naser, “A Rule-Based Expert System for Cybersecurity Threat Detection: Evolution, Applications, and the Hybrid AI Paradigm,” *International Journal of Academic Engineering Research*, vol. 9, no. 8, pp. 44–62, 2025.